Experiment:Decision Tree Classification using Python

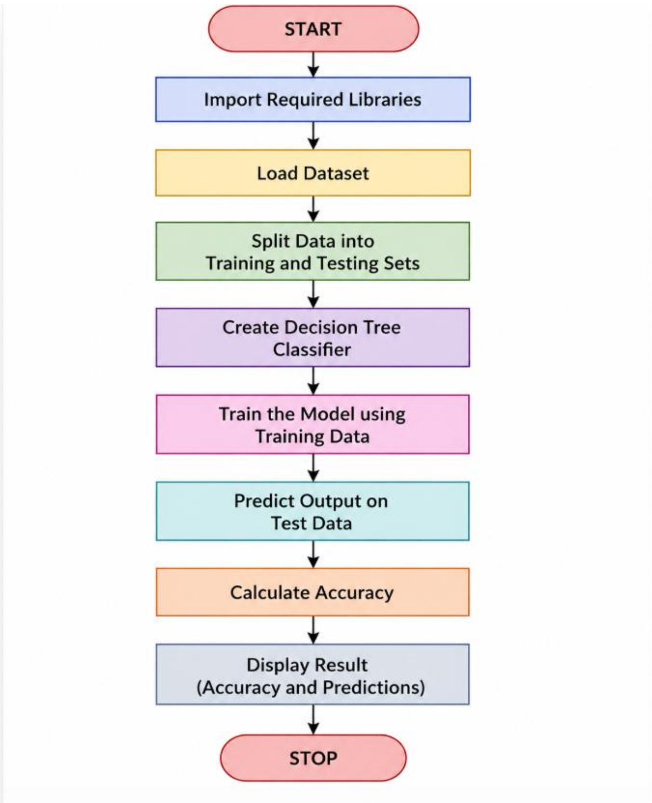
Aim

To implement a **Decision Tree Classifier** in Python using the Scikit-learn library and classify the given dataset accurately.

Algorithm:

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Create a Decision Tree Classifier.
5. Train the classifier using the training data.
6. Predict the output for the test data.
7. Calculate the accuracy of the model.
8. Display the prediction and accuracy.

FLOW CHART:



CODE:

```
# Decision Tree Classification

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Decision Tree model
model = DecisionTreeClassifier()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Actual Values:")
print(y_test)

print("\nPredicted Values:")
print(y_pred)

print("\nAccuracy =", accuracy * 100, "%")
```

OUTPUT:

```
Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy =
 100.0 %
Run completed in 7919.4000000059605ms
```

Result

The Decision Tree Classification algorithm was implemented successfully using Python. The model was trained on the Iris dataset and produced predictions with high accuracy.

Conclusion

Thus, the Decision Tree algorithm was implemented successfully using Python and Scikit-learn. The classifier learned the patterns in the training data and accurately classified the test samples, demonstrating that Decision Trees are simple, efficient, and effective for classification tasks.